

Task 4

```
cd/Users/emantaha/Desktop/Uvic/epigenomics/final/epigenomics/  
ATAC-seq
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ mkdir -p analyses  
(base) Emans-MacBook-Air:ATAC-seq emantaha$ mkdir -p data/  
bigBed.files  
(base) Emans-MacBook-Air:ATAC-seq emantaha$ mkdir -p data/bed.files  
(base) Emans-MacBook-Air:ATAC-seq emantaha$ mkdir -p analyses/  
peaks.analysis
```

```
# Download metadata (stomach , sigmoid colon ATAC-seq)
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ ../bin/  
download.metadata.sh "https://www.encodeproject.org/metadata/?  
type=Experiment&replicates.library.biosample.donor.uuid=d370683e-81e7-  
473f-8475-7716d027849b&status=released&assembly=GRCh38&biosample_ontol  
ogy.term_name=sigmoid+colon&biosample_ontology.term_name=stomach&assay  
_slims=DNA+accessibility"
```

```
# Get ATAC bigBed narrowPeak pseudoreplicated peaks ID
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ grep -F  
"bigBed_narrowPeak" metadata.tsv |\n  
grep -F "pseudoreplicated_peaks" |\n  
grep -F "GRCh38" |\n  
awk 'BEGIN{FS=OFS="\t"}{print $1, $11, $23}' |\n  
sort -k2,2 -k1,1r |\n  
sort -k2,2 -u > analyses/bigBed.peaks.ids.txt
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ cat analyses/  
bigBed.peaks.ids.txt
```

```
ENCFF287UHP  sigmoid_colon  
ENCFF762IFP  stomach
```

```
# Download bigBed files
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ cut -f1 analyses/  
bigBed.peaks.ids.txt |\n  
> while read filename; do  
> wget -P data/bigBed.files "https://www.encodeproject.org/files/  
$filename/@download/$filename.bigBed"
```

```
> done
```

check MD5

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ ../bin/selectRows.sh <(cut
-f1 analyses/bigBed.peaks.ids.txt) metadata.tsv |cut -f1,46 > data/
bigBed.files/md5sum.txt
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ cat data/bigBed.files/
md5sum.txt | \
> while read filename original_md5sum; do
> md5sum data/bigBed.files/"$filename".bigBed | \
> awk -v filename="$filename" -v original_md5sum="$original_md5sum" \
> 'BEGIN{FS=" "; OFS="\t"}{print filename, original_md5sum, $1}'
> done
```

```
# Make sure there are no files for which original and computed
MD5 hashes differ
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ awk '$2!=$3' data/
bigBed.files/md5sum.txt
```

#convert bigBed to BED

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ docker run --platform
linux/amd64 --rm \
> -v $PWD:$PWD -w $PWD \
> dgarrimar/epigenomics_course \
> bash -c "
> cut -f1 analyses/bigBed.peaks.ids.txt | \
> while read filename; do
> bigBedToBed data/bigBed.files/\${filename}.bigBed data/
bigBed.files/\${filename}.bed
> done
> "
```

```
total 29288
```

```
-rw-r--r--  1 emantaha  staff   7.4M Jun 11 14:22 ENCFF287UHP.bed
-rw-r--r--  1 emantaha  staff   6.9M Jun 11 14:22 ENCFF762IFP.bed
```

Number of ATAC-seq peaks intersecting promoter regions

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ cut -f-2 analyses/
bigBed.peaks.ids.txt | \
> while read filename tissue; do
> echo "=== $tissue: peaks overlapping promoters ==="
> bedtools intersect \
```

```
> -a data/bed.files/"$filename".bed \
> -b ../ChIP-seq/annotation/
genome.v24.protein.coding.non.redundant.TSS.bed \
> -u |\
> wc -l
> done
```

```
=== sigmoid_colon: peaks overlapping promoters ===
47871
=== stomach: peaks overlapping promoters ===
44749
```

Number of ATAC-seq peaks falling outside gene bodies(coordinates)

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ cut -f-2 analyses/
bigBed.peaks.ids.txt |\
> while read filename tissue; do
> echo "=== $tissue: peaks outside gene bodies ==="
> bedtools intersect \
> -a data/bed.files/"$filename".bed \
> -b ../ChIP-seq/annotation/
genome.v24.protein.coding.gene.body.bed \
> -v |\
> wc -l
> done
=== sigmoid_colon: peaks outside gene bodies ===
37035
=== stomach: peaks outside gene bodies ===
34537
```

Task 5

Task 1 : create regulatory_elements folder

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ mkdir -p /Users/emantaha/
Desktop/Uvic/epigenomics/regulatory_elements
```

```
(base) Emans-MacBook-Air:ATAC-seq emantaha$ cd /Users/emantaha/
Desktop/Uvic/epigenomics/final/epigenomics_uvic/regulatory_elements
```

Download metadata for H3K27ac and H3K4me1 for stomach , sigmoid colon

```
../bin/download.metadata.sh "https://www.encodeproject.org/metadata/?
type=Experiment&replicates.library.biosample.donor.uuid=d370683e-81e7-
473f-8475-7716d027849b&status=released&assembly=GRCh38&biosample_ontol
ogy.term_name=sigmoid+colon&biosample_ontology.term_name=stomach&assay
_slims=DNA+binding"
```

Get bigBed narrowPeak IDs for H3K27ac

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ grep -F
"H3K27ac" metadata.tsv |grep -F "bigBed_narrowPeak" |grep -F
"pseudoreplicated_peaks" |grep -F "GRCh38" |awk 'BEGIN{FS=OFS="\t"}
{print $1, $11, $23}' |sort -k2,2 -k1,1r |sort -k2,2 -u > analyses/
bigBed.H3K27ac.ids.txt
```

Get bigBed narrowPeak IDs for H3K4me1

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ grep -F
"H3K4me1" metadata.tsv |grep -F "bigBed_narrowPeak" |grep -F
"pseudoreplicated_peaks" |grep -F "GRCh38" |awk 'BEGIN{FS=OFS="\t"}
{print $1, $11, $23}' |sort -k2,2 -k1,1r |sort -k2,2 -u > analyses/
bigBed.H3K4me1.ids.txt
```

create folder for bed files

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ mkdir -p data/
bigBed.files data/bed.files
```

Download H3K27ac bigBed files

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ cut -f1
analyses/bigBed.H3K27ac.ids.txt |
> while read filename; do
>   wget -P data/bigBed.files/ "https://www.encodeproject.org/files/
$filename/@download/$filename.bigBed"
> done
```

Download H3K4me1 bigBed files

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ cut -f1
analyses/bigBed.H3K4me1.ids.txt |
> while read filename; do
>   wget -P data/bigBed.files/ "https://www.encodeproject.org/files/
$filename/@download/$filename.bigBed"
> done
```

Convert all bigBed to BED via Docker

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ docker run --
platform linux/amd64 --rm \
> -v $PWD:$PWD -w $PWD \
> dgarrimar/epigenomics_course \
> bash -c "
>   for f in data/bigBed.files/*.bigBed; do
>     base=$(basename "$f" .bigBed)
>     bigBedToBed "$f" data/bed.files/"${base}.bed"
>   done
> "
```

Task 2: Distal regulatory regions are usually found to be flanked by both H3K27ac and H3K4me1.

Overlap ATAC-seq distal peaks with histone marks

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ for tissue in
stomach sigmoid_colon; do
> H3K27ac_file=$(grep "$tissue" analyses/bigBed.H3K27ac.ids.txt | cut
-f1)
> H3K4me1_file=$(grep "$tissue" analyses/bigBed.H3K4me1.ids.txt | cut
-f1)
>
> echo "$tissue"
> echo "H3K27ac: $H3K27ac_file"
> echo "H3K4me1: $H3K4me1_file"
>
> bedtools intersect \
>   -a ../ATAC-seq/data/bed.files/"$(grep "$tissue" ../ATAC-seq/
analyses/bigBed.peaks.ids.txt | cut -f1).bed" \
>   -b data/bed.files/"$H3K27ac_file".bed \
>   -u | \
> bedtools intersect \
>   -a stdin \
>   -b data/bed.files/"$H3K4me1_file".bed \
>   -u > analyses/"$tissue".candidate.distal.regulatory.elements.bed
>
> echo "Number of candidate distal regulatory elements in $tissue:"
> wc -l analyses/"$tissue".candidate.distal.regulatory.elements.bed
>
> done
```

```
stomach
H3K27ac: ENCFF977LBD
H3K4me1: ENCFF844XRN
Number of candidate distal regulatory elements in stomach:
31927 analyses/stomach.candidate.distal.regulatory.elements.bed
sigmoid_colon
```

```
H3K27ac: ENCFF872UHN
H3K4me1: ENCFF724Z0F
Number of candidate distal regulatory elements in sigmoid_colon:
52377 analyses/
sigmoid_colon.candidate.distal.regulatory.elements.bed
```

Task 3: Focus on regulatory elements that are located on chromosome

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ for tissue in
stomach sigmoid_colon; do
> awk 'BEGIN{FS=OFS="\t"}$1=="chr1">{print $4, $2}' \
> analyses/"$tissue".candidate.distal.regulatory.elements.bed \
> > analyses/"$tissue".regulatory.elements.starts.tsv
>
> wc -l analyses/"$tissue".regulatory.elements.starts.tsv
> done

3546 analyses/stomach.regulatory.elements.starts.tsv
5413 analyses/sigmoid_colon.regulatory.elements.starts.tsv
```

Task 4: Focus on protein-coding genes located on chromosome 1.

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ awk
'BEGIN{FS=OFS="\t"}$1=="chr1"{
  if ($6=="+") {start=$2} else {start=$3};
  print $4, start
}' ../ChIP-seq/annotation/gencode.v24.protein.coding.gene.body.bed >
analyses/gene.starts.tsv
```

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ wc -l analyses/
gene.starts.tsv
2047 analyses/gene.starts.tsv
```

Task 5: python script

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ curl -k -L
-o ../bin/get.distance.py \
> "https://public-docs.crg.es/rguigo/Data/bborsari/UVIC/
epigenomics_course/get.distance.py"
```

```
# I have python 3 , i edited it inside python script
```

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ python3 ../bin/get.distance.py --input analyses/gene.starts.tsv --start 980000
```

```
ENSG00000187642.9    982093    2093
```

Task 6 Get closest gene for every regulatory element

```
for tissue in stomach sigmoid_colon; do
  cat analyses/"$tissue".regulatory.elements.starts.tsv | \
  while read element start; do
    python3 ../bin/get.distance.py \
      --input analyses/gene.starts.tsv \
      --start "$start"
  done > analyses/"$tissue".regulatoryElements.genes.distance.tsv
```

```
wc -l analyses/"$tissue".regulatoryElements.genes.distances.tsv
```

```
3546 analyses/stomach.regulatoryElements.genes.distances.tsv
5413 analyses/sigmoid_colon.regulatoryElements.genes.distances.tsv
```

Task 7 Compute mean and median distances

```
(base) Emans-MacBook-Air:regulatory_elements emantaha$ Rscript - <<
'EOF'
> for(tissue in c("stomach", "sigmoid_colon")) {
>   f <- paste0("analyses/", tissue,
".regulatoryElements.genes.distances.tsv")
>   df <- read.table(f, header=FALSE, sep="\t")
>   distances <- df[, 3]
>   cat(tissue, "\n")
>   cat(mean(distances), "\n")
>   cat(median(distances), "\n\n")
> }
> EOF
stomach
34227.18
12611

sigmoid_colon
45624.66
14608
```

